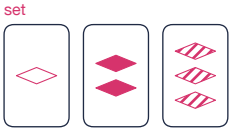
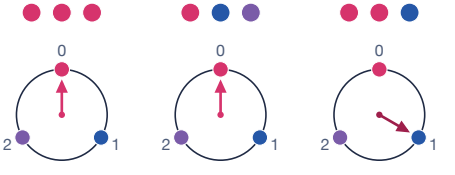
a A set, and three cards that are not one



number: all different
shape: all same
fill: all different
colour: all same

number: two alike: fails
shape: all same
fill: two alike: fails
colour: all same

b One attribute as a clock with three hours



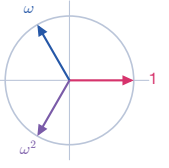
0 + 0 + 0 = 0
back to 0: a set

0 + 1 + 2 = 3
back to 0: a set

0 + 0 + 1 = 1
not back to 0

Colours as hours 0, 1, 2. Turn the hand once per card:
all same or all different brings it back to 0;
two alike never does.

c Hours on a circle



hour x is the point ω^x,

$\omega = e^{2\pi i/3}$;

its real part is 1 or $-\frac{1}{2}$